

DBMS Assignment 6

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1 Q1

(a) An attribute is extraneous if removing it does not change the closure of the FD set.

For LHS: In $\alpha \rightarrow \beta$, $A \in \alpha$ is extraneous if $(\alpha - A) \rightarrow \beta$ can be derived.

For RHS: In $\alpha \rightarrow \beta$, $A \in \beta$ is extraneous if $\alpha \rightarrow A$ still holds after removing it.

(b) $F = \{X \rightarrow YZ, Y \rightarrow XZ, Z \rightarrow XY\}$

In $X \rightarrow YZ$: If we remove Y , we get $X \rightarrow Z$. Since $Z \rightarrow XY$, we get $X \rightarrow Y$. So Y is extraneous.

If we remove Z , we get $X \rightarrow Y$. Since $Y \rightarrow XZ$, we get $X \rightarrow Z$. So Z is also extraneous.

But removing both together removes the dependency completely, so we must remove one at a time.

Q2

(a) Given $\alpha \rightarrow \beta$ and $\alpha \rightarrow \gamma$

$\alpha \rightarrow \beta$ $\alpha \rightarrow \gamma$ Using augmentation and transitivity, we get $\alpha \rightarrow \beta\gamma$

(b) Given $\alpha \rightarrow \beta$ and $\gamma\beta \rightarrow \delta$

$\alpha \rightarrow \beta$ $\alpha\gamma \rightarrow \gamma\beta$ $\gamma\beta \rightarrow \delta$ So $\alpha\gamma \rightarrow \delta$

Q3

(a) Assume relation is in 3NF but not in 2NF. Then there is a partial dependency $\gamma \rightarrow A$ where γ is part of key and A is non-prime.

This violates 3NF since γ is not a superkey and A is not prime. Contradiction. So 3NF implies 2NF.

(b) $R(\text{EmpID}, \text{DeptID}, \text{DeptName})$ FDs: $\text{EmpID} \rightarrow \text{DeptID}$, $\text{DeptID} \rightarrow \text{DeptName}$

Key is EmpID

No partial dependency, so 2NF holds. But there is transitive dependency, so not 3NF.

(c) $R(\text{StudentID}, \text{CourseID}, \text{StudentName})$ FD: $\text{StudentID} \rightarrow \text{StudentName}$

Key is $(\text{StudentID}, \text{CourseID})$

Partial dependency exists, so not 2NF.

Q4

If LHS has null values, decomposition causes issues.

Example: $R(\text{EmpID}, \text{DeptID}, \text{Manager})$ $\text{DeptID} \rightarrow \text{Manager}$

After decomposition: $R_1(DeptID, Manager)$ $R_2(EmpID, DeptID)$
 If DeptID is null, we cannot insert properly due to key constraints.

Q5

- (a) In BCNF, for relation (α, β) , α is key.
- (b) If foreign key not enforced, inconsistency occurs. Example: employee with invalid DeptID.
- (c) In 3NF decomposition, each $\alpha \rightarrow \beta$ gives relation with α as key. If key relation added, that key is primary.

Q6

$R = (A, B, C, D, E, G)$ $F = \{AB \rightarrow CD, ADE \rightarrow GDE, B \rightarrow GC, G \rightarrow DE\}$

- (a) $AB^+ = AB \rightarrow ABCG \rightarrow ABCDEG$

So AB is key.

- (b) After simplification: $F_c = \{B \rightarrow CG, G \rightarrow DE, ADE \rightarrow G\}$

- (c) Relations: (B, C, G) (G, D, E) (A, D, E, G)

Add key relation (A, B)

- (d) Final: $(B, C, G), (G, D, E), (A, D, E, G), (A, B)$

Q7

- (a) $AB^+ = ABCD \neq$ full relation So not superkey, violates BCNF
- (b) Decompose: (A, B, C, D) and (A, B, E, G)
 Then (A, B, C, D) splits into (B, D) and (A, B, C)
 Final: $(B, D), (A, B, C), (A, B, E, G)$
- (c) Not dependency preserving since $DE \rightarrow B$ lost

Q8

- (a) Repetition of data causes update anomaly Example: DeptBudget repeated
- (b) Cannot insert dept without course (insertion anomaly)

Q9

Rule is wrong.

Example: $A \rightarrow B, C \rightarrow B$ does not imply $A \rightarrow C$

Correct rule: $A \rightarrow B, B \rightarrow C$ implies $A \rightarrow C$

Q10

- (a) $CD \rightarrow AB$ $A \rightarrow BC$
 So $CD \rightarrow E$ using transitivity
- (b) Decompose: (A, B, C) and (A, D, E, G)
 Then (A, D, E) and (A, D, G)
 Final: $(A, B, C), (A, D, E), (A, D, G)$

- (c) Lossless: yes
- (d) Not dependency preserving